6(a)	Threshold frequency refers to the minimum frequency of the illuminating electromagnetic radiation that will cause a photoelectron to be ejected for a particular metal.	[1]
6(b)(i)	Energy of a photon, $E = \dfrac{hc}{\lambda} = \dfrac{\left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)}{450\times10^{-9}} = 4.42\times10^{-19}\ \text{J}$	[1] sub [1] ans
6(b)(ii)	Power incident on metal, $P = (2.7 \times 10^{3})(3.0 \times 10^{-4}) = 0.81$ W $P = \left(\dfrac{N}{t}\right)E$ $\Rightarrow \dfrac{N}{t} = \dfrac{P}{E} = \dfrac{0.81}{4.42\times10^{-19}} = 1.83\times10^{18}\ \text{s}^{-1}$	[1] value [1] ans

6(b)(iii)	Max. K.E. = eV_s = $(1.6 \times 10^{-19})(1.6) = 2.56 \times 10^{-19}$ J Applying Einstein Photoelectric equation, Work function, $\phi = hf - \text{max. K.E.} = 4.42\times10^{-19} - 2.56\times10^{-19} = 1.86\times10^{-19}$ J Threshold wavelength, $\lambda = \dfrac{hc}{\phi} = \dfrac{(6.63\times10^{-34})(3.00\times10^{8})}{1.86\times10^{-19}} = 1.07\times10^{-6}$ m	[1] value [1] value [1] ans
6(c)(i)	From the graph, <u>λ_{min} is the same for both spectra</u>. $eV = \dfrac{hc}{\lambda_{min}} \Rightarrow V = \dfrac{hc}{e\lambda_{min}}$	[1]
6(c)(ii)	From the graph, $\lambda_{min} = 16\times10^{-12}$ m $V = \dfrac{hc}{e\lambda_{min}} = \dfrac{6.63\times10^{-34}\times3.00\times10^{8}}{1.60\times10^{-19}\times16\times10^{-12}} = 7.8\times10^{4}$ V	[1] sub [1] ans
7(a)(i)	The half-life of a radioactive nuclide is the <u>time taken for the</u>	